After meeting with the clients, we determined that the main purpose of the app rating prediction system is to give them a clearer understanding of how their app is performing. Their primary goals are to build on the current system and add enhancements that improve overall usability. Each person had their own requests, which helped clarify how we should proceed with the upcoming project.

The users of this system will be quality analysts and customer support analysts, while the stakeholders include management and higher-level executives. The current system is lacking in several key areas. It provides only basic measurements, such as the average review, and does not account for review explanations, treating all ratings the same. It lacks a strong user interface and dashboard functions that support ease of use for non-data users. It also lacks an automated updating system, requiring users to refresh the data manually.

To address the first issue, we plan to create a review categorization system that prioritizes reviews offering insight into why the app received its rating. We also plan to add a feature that may be able to predict user reviews based on system errors, for example, if a system goes down, we can assume user reviews will mention system failure. To address the second issue, we plan to create a new dashboard that will allow users to filter reviews based on text and rating. We also want this dashboard to support filtering by sentiment, performance, and time. It will include a simple user interface that enables non-data users to understand general statistics. Finally, we plan to add a nightly refresh to the app reviews. A scraper will collect new reviews and process them, such as identifying key sentiments.

One copy of this new data will be stored, and another will be sent through the app review system. Because this data is intra-organizational, it will require work emails to access it.

With this new system, the stakeholders' requirements will be met. Quality analysts will be able to assess the app not just through a simple average rating but through more in-depth metrics. Customer success teams will be able to better address customer issues by using predictive reviews and identifying common complaints. Management will also have an easier time visualizing and examining app performance through an easy-to-use interface that provides clear, readable metrics.

Our success metrics will be based on several key features of the system. The first, and simplest, will be user feedback on the dashboard. This will ensure that the interface meets the standards and expectations of the clients. We will also measure the success of the data-processing component by evaluating how accurately it categorizes meaningful reviews. To do this, we will feed the system a test dataset containing a variety of reviews that address different issues and assess whether the system labels them correctly. Finally, we will test the effectiveness of the predictive model by simulating an app outage and determining whether the system correctly predicts that new reviews will mention the outage.